Section 2

Q6 Reg_Assertion Tb

```
# Allocation: Simulator allocated 4674 kB (elbread=427 elab2=4110 kernel=137 sdf
# KERNEL: ASDB file was created in location /home/runner/dataset.asdb
# KERNEL: Pass: q=00000000, expected=00000000, reset=1, load=0, d=00000000
# KERNEL: Pass: q=00000000, expected=00000000, reset=1, load=0, d=00000000
# KERNEL: Pass: q=00000000, expected=00000000, reset=0, load=0, d=00000000
# KERNEL: Pass: q=1234abcd, expected=1234abcd, reset=0, load=1, d=1234abcd
# KERNEL: Pass: q=1234abcd, expected=1234abcd, reset=0, load=0, d=1234abcd
# KERNEL: Pass: q=1234abcd, expected=1234abcd, reset=0, load=0, d=ffff0000
# KERNEL: Pass: q=1234abcd, expected=1234abcd, reset=0, load=0, d=ffff0000
# KERNEL: Pass: q=1234abcd, expected=1234abcd, reset=0, load=0, d=ffff0000
# KERNEL: Pass: q=deadbeef, expected=deadbeef, reset=0, load=1, d=deadbeef
# KERNEL: Pass: q=deadbeef, expected=deadbeef, reset=0, load=0, d=deadbeef
# KERNEL: Pass: q=deadbeef, expected=deadbeef, reset=0, load=0, d=deadbeef
# KERNEL: Test completed successfully!
# RUNTIME: Info: RUNTIME_0068 testbench.sv (82): $finish called.
# KERNEL: Time: 110 ns,  Iteration: 0,  Instance: /reg32_tb,  Process: @INITIAL#
# KERNEL: stopped at time: 110 ns
```

Q7_cntr

Log   Share

```
# KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.
# KERNEL: SLP simulation initialization done - time: 0.0 [s].
# KERNEL: Kernel process initialization done.
# Allocation: Simulator allocated 5596 kB (elbread=459 elab2=4967 kernel=169 sdf=0)
# KERNEL: ASDB file was created in location /home/runner/dataset.asdb
# KERNEL: [TXN] up_dn=0
# KERNEL: [MON] count=0
# KERNEL: [SCB] PASS actual=0 expected=0
# KERNEL: [TXN] up_dn=1
# KERNEL: [MON] count=1
# KERNEL: [SCB] PASS actual=1 expected=1
# KERNEL: [TXN] up_dn=0
# KERNEL: [MON] count=0
# KERNEL: [SCB] PASS actual=0 expected=0
# KERNEL: [TXN] up_dn=0
# KERNEL: [MON] count=1
# KERNEL: [SCB] PASS actual=1 expected=1
# RUNTIME: Info: RUNTIME_0068 testbench.sv (235): $finish called.
# KERNEL: Time: 220 ns,  Iteration: 0,  Instance: /tb_top,  Process: @INITIAL#232_2@.
# KERNEL: stopped at time: 220 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
# VSIM: Simulation has finished.
Done
```

Shift_Reg

```
# KERNEL: Kernel process initialization done.
# Allocation: Simulator allocated 5600 kB (elbread=459 elab2=4971 kernel=169 sdf=0
# KERNEL: ASDB file was created in location /home/runner/dataset.asdb
# KERNEL: [TXN] shift_en=0 dir=1 d_in=0
# KERNEL: [MON] q_out=00000000
# KERNEL: [SCB] PASS actual=00000000 expected=00000000
# KERNEL: [TXN] shift_en=1 dir=0 d_in=1
# KERNEL: [MON] q_out=00000000
# KERNEL: [SCB] PASS actual=00000000 expected=00000000
# KERNEL: [TXN] shift_en=0 dir=1 d_in=1
# KERNEL: [MON] q_out=00000001
# KERNEL: [SCB] PASS actual=00000001 expected=00000001
# KERNEL: [TXN] shift_en=0 dir=1 d_in=0
# KERNEL: [MON] q_out=00000001
# KERNEL: [SCB] PASS actual=00000001 expected=00000001
# KERNEL: [TXN] shift_en=1 dir=0 d_in=0
# KERNEL: [MON] q_out=00000001
# KERNEL: [SCB] PASS actual=00000001 expected=00000001
# KERNEL: [TXN] shift_en=0 dir=1 d_in=0
# KERNEL: [MON] q_out=00000010
# KERNEL: [SCB] PASS actual=00000010 expected=00000010
# RUNTIME: Info: RUNTIME_0068 testbench.sv (221): $finish called.
# KERNEL: Time: 320 ns,  Iteration: 0,  Instance: /tb_top,  Process: @INITIAL#218_
# KERNEL: stopped at time: 320 ns
```

Multiplier

```
# Allocation: Simulator allocated 5505 kB (elbread=
# KERNEL: ASDB file was created in location /home/r
# KERNEL: Output = 0
# KERNEL: Output = 0
# KERNEL: Output = 26656
# KERNEL: Output = 26656
# KERNEL: Output = 14480
# KERNEL: Output = 14480
# KERNEL: Output = 10800
# KERNEL: Output = 10800
# KERNEL: Output = 16100
# KERNEL: Output = 16100
# KERNEL: Output = 17248
# KERNEL: Output = 17248
# KERNEL: Output = 6579
# KERNEL: Output = 6579
# KERNEL: Output = 952
# KERNEL: Output = 952
# KERNEL: Output = 19764
# KERNEL: Output = 19764
# KERNEL: Output = 11160
# KERNEL: Output = 11160
# KERNEL: Output = 1976
# KERNEL: Output = 1976
```